EXPRIMENT-13

13.) Known Plaintext Attack on Hill Cipher

Aim:

To write a Python program to recover the Hill cipher key using known plaintext and ciphertext pairs.

PROGRAM:



```
def inverse(a, m):
    for i in range(1, m):
        if (a*i) % m == 1:
            return i
    return None


def multiply(A, B):
    result = [[0, 0], [0, 0]]

    for i in range(2):
        for j in range(2):
            for k in range(2):
                result[i][j] += A[i][k] * B[k][j]

            result[i][j] %= 26

    return result


def matrix_inverse(M):
    a, b = M[0]
```



```
main.py

    det = (a*d - b*c) % 26
    inv = inverse(det, 26)

    return [
        [(inv*d) % 26, (-inv*b) % 26],
        [(-inv*c) % 26, (inv*a) % 26]
    ]


P = [[7, 11], [4, 15]]

K = [[3, 3], [2, 5]]

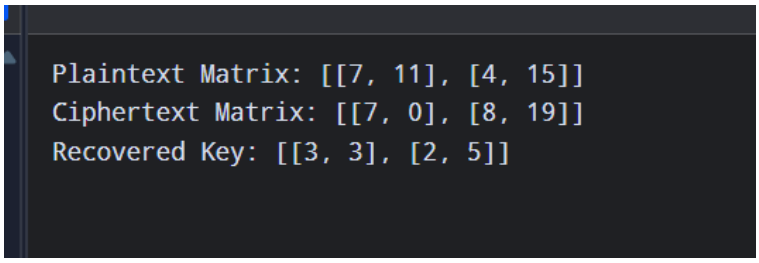
C = multiply(K, P)

P_inv = matrix_inverse(P)

recovered_key = multiply(C, P_inv)

print("Plaintext Matrix:", P)
print("Ciphertext Matrix:", C)
print("Recovered Key:", recovered_key)
```

OUTPUT:



```
Plaintext Matrix: [[7, 11], [4, 15]]
Ciphertext Matrix: [[7, 0], [8, 19]]
Recovered Key: [[3, 3], [2, 5]]
```

RESULT:

Thus, the original Hill cipher key was successfully recovered using a known plaintext attack.